PAS FINAL EXAM PAST PAPER

Descriptive Statistics

Question 01

Consider the following set of execution times (in milliseconds) for different algorithms on various datasets.

Dataset:

26, 32, 45, 32, 24, 29, 23, 33, 43, 37, 15, 32, 22, 52, 27, 19, 24, 35, 43, 32, 41, 35, 19, 32, 34, 29, 37, 34, 34, 49, 23

Type your answers within the given spaces except for Part 3.

1. Find the following values:

a. Mode
b. Q1 (First quartile)
c. Median
d. Q3 (Third quartile)
e. Mean
f. IQR

2. Analysis of Outliers

a) Upper boundary to check outliers
b) Lower boundary to check outliers
c) How many outliers are there in the dataset?

3. Conceptual Question

Which of the following statistics would remain unchanged if a constant value was added to every execution time in the dataset?

Question 02

Consider the variations in response times observed when a mobile application retrieves fitness tracking data from different cloud-hosted APIs.

Data:

12, 18, 27, 23, 18, 14, 17, 25, 34, 33, 16, 27, 31, 39, 35, 26, 28, 37, 40, 42

30, 36, 38, 24, 41, 13, 15, 14, 23, 43, 18

1. Find the following values.

a. Mode
b. Q1
c. Median
d. Q3
e. Mean
f. IQR

2. Analysis of outliers

a) Upper boundary to check outliers
b) Lower boundary to check outliers
c) How many outliers are there in the data set?

3.According to the box plot of the data set, what is the shape of the distribution?

Normal Distribution

Question 01

Most graduate schools of computer science require applicants to take a Programming Aptitude Test (PAT). Scores on the PAT are approximately normally distributed with a mean of 680 and a standard deviation of 75.

Find the answers to the following questions (Type the answer in the given space).

1) What is the probability of an individual scoring above 700 on the PAT?

[ ] (Keep all decimal places in the answer)

2) How high must an individual score on the PAT in order to score in the highest 2%?

[ ] (Round off the answer up to 2 decimal places)

3) What is the probability of an individual scoring below 600 on the PAT?

[ ] (Keep all decimal places in the answer)

Question 02

A researcher found that pulse rates of women, who are affected by COVID-19 virus has a mean of 75 and the standard deviation is 8.

Assuming that pulse rate of women follows a normal distribution, find the following probabilities.

1. Probability that a COVID-19 infected woman has less than 70 pulse rate (Keep all the decimal places in the answer)

2. Probability that a COVID-19 infected woman has pulse rate in between 55 and 90 (Keep all the decimal places in the answer)

3. At which pulse rate, 30% of COVID-19 infected women have more than that pulse rate? (Keep the answer with two decimal points)

Question 03

A researcher has found that average time taken to load a certain educational website by your internet service provider is 3 minutes with a standard deviation of 1.3 minutes. Assuming that this time follows a normal distribution, find the following probabilities.

1. Probability that time taken to load the above page is at least 4 minute (Keep all the decimal places in the answer)

2. Probability that 2 to 5 minutes have taken to load the above page (Keep all the decimal places in the answer)

3. What is the time where 52% of service providers took more than this time to load the page? (Keep the answer with two decimal points)

Statistical Inference -Confidence Interval

Question 01

A company monitors the average response time of its cloud servers to ensure optimal performance for users. The response time is measured in seconds, and the company wants to estimate the true mean response time of all servers. A random sample of 25 servers shows an average response time of 1.75 seconds with a standard deviation of 0.25 seconds. Construct a 95% confidence interval for the true mean response time.

1. Lower limit equation:

2. Upper limit equation:

3. Critical value:

4. Lower limit value:

5. Upper limit value:

6. Let's assume that lower limit is a and upper limit is b. Interpret the confidence interval

a) We are 95% confident that the true mean response time will be between a and b
b) We are 5% confident that the true mean response time will be between a and b
c) We are 2.5% confident that the true mean response time will not be between a and b
d) We are 95% confident that the true mean response time will not be between a and b.
e) We are 2.5% confident that the true mean response time will be between a and b.

Question 02

A university wants to estimate the average time students spend completing an online examination. A random sample of 36 students has a mean completion time of 80 minutes and a standard deviation of 12 minutes.

Construct a 95% confidence interval for the population mean completion time.

1. Write the lower limit equation.
2. Write the upper limit equation.
3. Find the critical value.
4. Calculate the lower limit.
5. Calculate the upper limit.
6. Interpret the confidence interval.

Question 03

A tech firm wants to estimate the average time (in seconds) it takes for its database server to execute a query.

A sample of 60 queries shows an average execution time of 0.72 seconds with a population standard deviation of 0.08 seconds. Construct a 95% confidence interval for the true mean execution time of the database server.

1. Lower limit equation

2. Upper limit equation

3. Critical value

4. Lower limit value

5. Upper limit value

6. Interpretation of the confidence interval

Let's assume that lower limit is a and upper limit is b.

Select the correct interpretation:

a) We are 95% confident that the true mean execution time of the database server will be between a and b.
b) We are 95% confident that the true mean execution time of the database server will not be between a and b.
c) We are 2.5% confident that the true mean execution time of the database server will not be between a and b.
d) We are 2.5% confident that the true mean execution time of the database server will be between a and b.
e) We are 5% confident that the true mean execution time of the database server will be between a and b.

Hypothesis Testing

Question 01

A software company claims that its new algorithm reduces the average response time of its server from the current average of 250 milliseconds.

A random sample of 40 server response times after implementing the new algorithm has a mean response time of 245 milliseconds with a standard deviation of 18 milliseconds.

The company wants to test whether the new algorithm has significantly reduced the average server response time than 250 milliseconds.

Test the company's claim at a 5% level of significance.

Answer the following questions:

1. What is the hypothesis?
2. What is the distribution of the test statistic?
3. What is the critical value? (Round off the answer to two decimal places.)
4. Select the correct rejection region for the test.
5. What is the test statistic value? (Round off the answer to four decimal places.)
6. Give the scientific conclusion.
7. What is the general conclusion?

Question 02

A software development team claimed that the new API endpoint do not change the true average response time from 150 milliseconds (which is the current response time).A random sample of 45 requests recorded a mean response time of 156 milliseconds with population standard deviation of 12 milliseconds.

Test the claim at 5% level of significance.

1. What is the hypothesis?

a) $H0: \mu = 150$ vs $H1: \mu \neq 150$
b) $H0: \mu \geq 150$ vs $H1: \mu < 150$
c) $H0: \mu \geq 150$ vs $H1: \mu \neq 150$
d) $H0: \mu \leq 150$ vs $H1: \mu \neq 150$
e) $H0: \mu \leq 150$ vs $H1: \mu > 150$

2. What is the distribution?

3. What is the critical value?

4. Select correct rejection region.

5. What is the test statistic value?

6. Give the scientific conclusion.

7. What is the general conclusion?

Chi-Square Test

A school administrator wants to know if there is an association between students' academic year (1st year, 2nd year, 3rd year) and their preferred mode of learning (Face-to-face, Online, Hybrid).

He collected data from 140 students and the results are shown below.

Academic Year	Face-to-Face	Online	Hybrid	Total
1st Year	15	13	12	40
2nd Year	14	16	10	40
3rd Year	18	20	22	60
Total	47	49	44	140

Test at 5% significance level whether there is a significant relationship between students' academic year and their preferred mode of learning.

Chi-Square Test Statistic

$$X^2 = \sum_i \sum_j \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

1. Hypothesis

Select the correct hypothesis.

a) H0: Students' academic year and their preferred mode of learning are not independent. Vs. H1: Students' academic year and their preferred mode of learning are independent.
b) H0: Students' academic year and face-to-face mode of learning are independent. Vs. H1: Students' academic year and face-to-face mode of learning are not independent.
c) H0: Students' academic year and their preferred mode of learning are independent. Vs. H1: Students' academic year and their preferred mode of learning are not independent.
d) H0: Students' academic year and online mode of learning are independent. Vs. H1: Students' academic year and online mode of learning are not independent.